



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

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TECHNICAL SPECIFICATION

FOR

11kV & 22kV, (3) POLE AIR DISCONNECTOR SWITCH

(200,400,630A)



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Issue: NOV-2020 / Rev- 1

- توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.
- يلزم ارفاق المواصفات الفنية للمصهرات جهد متوسط (EDMS 19-402/EDMS 19-400) في حاله السكاكين الهوائية سعة ٢٠٠ أمبير.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

CONTENTS

<u>Clause</u>	<u>Description</u>	<u>Page</u>
1	SCOPE	3
2	APPLICABLE STANDARDS	3
3	ENVIRONMENTAL CONDITIONS	4
4	OPERATING MECHANISM	4
5	DESCRIPTION	5
6	CONTACT MATERIAL	5
7	TERMINAL	5
8	ARCING HORNS	6
9	EARTHING SWITCHES	6
10	RATINGS OF FUSE SUPPORTING INSULATOR	6
11	GENERAL REQUIREMENTS/CONSTRUCTIONAL DETAILS	7
12	THICKNESS OF GALVANIZING COATING	7
13	FUSES	7
14	TECHNICAL DATA	8
15	TESTS	9
16	DRAWINGS AND CATALOGUES	10
17	MARKING	10
18	GENERAL NOTES	11
-	GURANTEE TABLE 1	12
-	GURANTEE TABLE 2	14



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

1. SCOPE

This specification covers the minimum requirements of design, manufacture, testing, supply, construction and delivery of 11kV & 22 kV, (3) pole, alternating current disconnectors with earthing switches, designed for outdoor and open terminal installations for service frequency 50 Hz. Three pole disconnectors equipped with earth switch for all rated and expulsion fuse cut-out or H.R.C fuses for 200 A only. The outdoor, medium voltage three-pole disconnector should be suitable for (on / off / earth) switching of power lines at no-load and at outdoor applications, they will be installed in horizontal or vertical position on a metallic frame.

2. APPLICABLE STANDARDS

Unless otherwise specified, the disconnector switch should comply with the latest edition of IEC publications or any other standard at the same level or above related to the subject of specification.

S.N	Standard No.	Description
1	IEC 62271-103	High-voltage switchgear and control gear - Part 103: Switches for rated voltages above 1 kV up to and including 52 kV.
2	IEC 62271-102	High-voltage switchgear and control gear – Part 102: Alternating current disconnectors and earthing switches.
3	IEC 62271-1	High-voltage switchgear and control gear – Part 1: Common specifications.
4	ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles — Specifications and test methods.
5	DIN 50976	Protection against corrosion; hot galvanizing (general galvanizing); requirements and tests.
6	ES 863	The coating of galvanizing with zinc metal on iron and steel works by hot immersion.
7	IEC 60815	Selection and dimensioning of high voltage insulators intended for use in polluted conditions-part1: definitions, information and general principal.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

8	IEC 60507	Artificial pollution tests on high voltage insulators to be used on AC system.
9	IEC 60282-1	High voltage fuse - part1 current limiting fuse
10	IEC 60282-2	High voltage fuse - part2 expulsion fuse
11	BS-3953	Specification for synthetic resin bonded woven glass fabric laminated sheet.
12	IEC62271-1:2017	Common Specification for alternating current switchgear and controlgear

3. ENVIRONMENTAL CONDITIONS

The performance of the 3-pole disconnector components should be suitable for Egypt environmental conditions:

- Ambient temperatures : 45°C (50 °C as option).
And polluted water is possible. The sunlight is very intense and the temperature of exposed 70 °C.
- The maximum solar radiation intensity on a horizontal surface: 1000 w/m².
- Severe surface contamination from salt, dust, heavy industrial pollution, sandstorms, fine dust often carried in suspension in the atmosphere even on calm days, can be expected.
- Altitude less than : 1000 meters
- Relative humidity : 10 - 95 %.
- Pollution : Max. class D (heavy)

4. OPERATING MECHANISM

The 3-pole disconnector should be designed to secure isolator mechanically in close, open and earth positions. The whole set should be complete with an operating pipe of 6 meter length and three piceces of 19 mm diameter. (with length suiting mechanism mounted 1.5 m above ground level at pole foot), turn-buckle type handle has a thickness not less than 5 mm with clevis pad-lock and locking



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

arrangement, (on/off/earth) indicator, intermediate guide levers, and all necessary clamps and bolts for fixing these on a pole. safety insulators between pipes have to be installed.

5. DESCRIPTION

a-The 3-pole disconnecter should consist of three gang operated single pole isolators coupled together by one square sections of galvanized steel. Each single pole unit consists of at least 2 insulators of rated voltage 11 & 22 kV. One fixed and one movable post insulator. Stainless steel pins are used for hinges. Insulators should be of porcelain preferably brown or high resistance outdoor epoxy resin, Silicon rubber or hydrophobic-cycloaliphatic

b- Each of the three poles should be mounted independently on the common base. Phase distance between poles should be regulated and then finally interconnected with common shaft to ensure simultaneous operation of all three phases.

6. CONTACT MATERIAL

Self-aligning fixed contact and moving blade are of hard drawn high conductivity flat bar section copper silver-plated over contact area. Thickness of silver plating should be minimum 8 μm . Spiral springs made from stainless steel can be fitted at both sides of fixed contact to maintain proper contact pressure.

7. TERMINAL:

The terminal should be supplied with compression clamps type suitable for ACSR 150/25 mm², 70/12 mm², 240/40 mm² and for AAAC 300 mm².

The connecting surface should be made of the electrical corrosion proof (bi-metallic corrosion proof) Flexible conductor cables: to complete the current path from the moving blades to the Bing end.

They are manufactured from fine stranded copper wires which ensure maximum flexibility and insulated with PVC sheath.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

8. ARCING HORNS

- Arcing horns should consist of two steel pins. One fixed at fixed contact act (blade-socket) which should be stationary. The other fixed at contact blade and should be movable.
- Arcing spring whip should be installed on the fixed contact, on the moving contact side charging element has to be installed. arcing horns or spring whips has to be made from stainless steel
- Horns or whips should be capable of dealing with breaking load up to 20 A at 0.7 P.F and reactive breaking current of 5 A at 0.15 p.f. Load breaking have to be independent from operating speed - performed by human or motor. Load breaking capability has to be confirmed by the tests in the independent laboratory and at least 100 breaking operation has to be guaranteed

9. EARTHING SWITCHES

- The disconnecter should equip with one earthing switch for 200A rating at load side, and one or two earthing switch (as option according to ...EDC requirements) for 400A and 630A ratings, and should have the possibility to install earthing switches from both sides of switch disconnecter.
- Earthing switch should be operated manually and a mechanical interlock between switch disconnecter and earthing switch has to be provided.

10. RATINGS OF FUSE SUPPORTING INSULATOR (in case of IN LINE INSULATOR FOR 200 A DISCONNECTOR):

- Height of half insulator ≥ 22 cm.
- Min. leakage path of half insulator ≥ 50 cm.
- Isolator min. rated short time current withstand 16 KA for 1 s, 8 KA For 3 s.
- Alt. Voltage withstand for 1 min (dry) 50 kV
- Alt. Voltage withstand for 1 min (wet) 30 kV
- Dry flashover v. ~ 60 cm
- Wet flashover v. > 45 kV
- Impulse withstand V. (1.2/50 μ s) 15 + ve, 15 - ve impulses 125 kV
- Bending force 750kN



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

11. GENERAL REQUIREMENTS/CONSTRUCTIONAL DETAILS :

- In case of drop out Fuses the cutouts should be of single vent type (downward) having a front connected fuse carrier suitable for angle mounting.
- All ferrous parts should be hot dip galvanized. Nut, bolts and spring washers should be electro-galvanized.

12. FUSES:

Fuses should be compatible with latest version of (According toEDC requirements.):

(EDMS19-402) technical specification for drop out Fuses (D.O.F)

OR

(EDMS19-400) technical specification of High Rupture Current Fuse

13. All galvanized parts should be the thickness of galvanizing coating according to the following table:-

Type of workpieces	Local coating thickness , μm	Corresponding mass per unit area , g/m^2
Steel parts less than 1 mm thick	50	360
Steel parts 1 mm to under 3 mm thick	55	400
Steel parts 3 mm to under 6 mm thick	70	500
Steel parts 6 mm and thicker	85	610



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

14. TECHNICAL DATA:

TECNECAL DATA		11 kV	22 kV
1	System rated voltage (Nominal voltage)	11 kV	22 kV
2	System highest voltage	12 kV	24 kV
3	Disconnecter rated current @40°C	200 - 400 - 630 A	
4	Disconnecter min. rated short time current withstand for 1 s.	16 KA (1 s) , 8 KA For (3 s)	
5	Disconnecter peak withstand current	40 kA	
6	Pole centers spacing	510 mm	560 mm
7	Least mechanical endurance	2000 c/o cycles	
8	Rated frequency	50 Hz	
9	Min creepage distance	4 cm/ kV	<u>4</u> cm/ kV
10	One-minute dry power frequency withstand voltage (r.m.s)		
	Phase – earth	28 kV	50 kV
	Phase – phase	28 kV	50 kV
	Through open contact	32 kV	60 kV
11	Dry impulse withstand voltage		
	Phase – earth	75 kV	125 kV
	Phase – phase	75 kV	125 kV
	Through open contact	85 kV	145 kV
12	Contact resistant	not more than <u>60</u> μΩ	
13	Pollution factor	not less than 50 Micro	
14	Copper contact with density 1.5A/mm, purity 99.999% with 2 holes Ø 13		



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

15. TESTS:

ROUTINE TESTS according to relevant IEC IEC62271-102 & as following:

- a) Visual inspection:
 - Dimension.
 - Creepage distance
 - Base of isolator switch hot dip galvanized and with tall and thickness (not less than 130 cm, 5mm thickness for C-Shape frame or 3 mm thickness for closed profile frame).
- b) Measuring of silver plate thickness and galvanizing coating thickness .
- c) Measuring of the resistance of the main circuit
- d) Dielectrical test on the main circuit
- e) Mechanical operating tests
- f) Verification of earthing function

TYPE TESTS:

according to relevant IEC62271-102 & as following:

- a) The following tests should be carried out at works in accordance with IEC standards:
 - Power frequency voltage withstand test for 1 min when knife closed at 50 Hz (dry & wet).
 - Power frequency voltage withstand test for 1 min when knife open at 50 Hz (dry & wet).
- b) Temperature rise test: using nominal current between contact poles (provide it doesn't exceed 65 °C) to measure the contact resistance before and after test.
- c) Impulse withstand voltage test (1.2/50 μ s).
 - On switch knife's poles closed
 - On switch knife's poles open
- d) Maximum conductivity measurement at operating voltage for polluted insulators.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

- e) Short time withstand current and peak withstand current tests
- f) Test to verify the short-circuit making performance of earthing switches
- g) Mechanical tests
- h) Test on fiber glass tube in case of 200A disconnecter.

One sample of fiber glass tube will be selected from the lot under inspection and also for the type test and tested for the following tests as per BS-3953/105. Grade MF4 with latest amendment if any.

- 1- Electric strength of 25kV when tested edgewise in oil at 90°C.
- 2- Insulation resistance of 10 meg. Ohm (minimum) after emersion in water at 500V DC.

16.D RAWINGS AND CATALOUGES

The tender should submit technical catalogues and detailed dimensional drawings (if it is not submitted with the tender) for approval by ...EDC before manufacturing starts or before ordering for importing by the contractor. ...EDC reserves the right to require revision of the manufacturer's drawings without change in contract price, provided the general scope of work is not substantially altered.

17.MARKING:

The following data should be indelibly marked on disconnecter on metal nameplate engraved or laser printed on the disconnecter body, and located in a place such that they are visible:

- The Manufacturer's Name.
- Serial Number and Year of Manufacturing.
- Applicable Standard.
- Rated Frequency.
- Rated current @40 °C.
- Nominal voltage.
- Disconnecter peak withstand current.
- One-minute dry power frequency withstands voltage (r.m.s).
- Dry impulse withstands voltage.
- Distribution company name.



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EDMS 19-403-1

Date: 23-11-2020

ملحوظة هامة:

- يلزم إرفاق كتالوج الشركة المصممة للسكاكين والتي قامت بإجراء الإختبارات النوعية عليها بالعرض والتي تم بناء على هذا التصميم تصنيع هذه السكاكين.
- يلزم تغليف جميع السكاكين بالبلاستيك لحماية عوازل الجهد المتوسط ونقاط التلامس من الأتربة.
- يلزم دهان عوازل السكاكين بمادة RTV في حالة العوازل البورسلين .
- يلزم تقديم نسخة الكترونية بها جميع البيانات الفنية والرسومات والكتالوجات الخاصة بالسكينة.
- بالنسبة للسكاكين حمل ٢٠٠ امبير يتم استخدام كابلات شعربعدد ٢ كابل قطاع ٥٠X١ مم^٢ والسكاكين حمل ٤٠٠ و ٦٣٠ امبير كابلات شعر بعدد ٣ كابل قطاع ٥٠X١ مم^٢



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

GUARANTEE TABLE 1

1- PARTICULAR	
• Maker's name	
• Type	
• Number of poles	
• Type of insulators	
• Dimension of insulators	
• Minimum height	Mm
• Minimum diameter	Mm
• Type of contacts	
• Material of contacts	
• Contact surface area	mm ²
• Blades : dimension , material	
• Contact resistance	$\mu\Omega$
• Thickness of silver	μm
2- PERFORMANCE	
• Rated voltage	kV
• Nominal current at 40 °C	A
• Disconnecter min. rated short time current withstand for 1 s/3s.	kA
• Disconnecter peak withstand current	kA
• Pole centers spacing	Mm
• Least mechanical endurance at least	c/o cycles
• Rated frequency	HZ
• Min creepage distance	Cm/kV



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

• One-minute dry power frequency withstand voltage (r.m.s) Phase – earth	kV
• One-minute dry power frequency withstand voltage (r.m.s) Phase – phase	kV
• One-minute dry power frequency withstand voltage (r.m.s) Through open contact	kV
• Dry impulse withstand voltage Phase – earth	kV
• Dry impulse withstand voltage Phase – phase	kV
• Dry impulse withstand voltage Through open contact	kV
• Pollution factor not less than 50 Micro	Yes/no
• Hot Deep galvanizing steel as per IEC and according to sheet steel thickness	Yes/no

I/ We guarantee the technical data given above for the offered load break switch.

Signature:

Date:



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 19-403-1

Date: 23-11-2020

GUARANTEE TABLE 2 (for 200A disconnectors)

Fuse supporting insulator (in case of in line insulator) ratings	kV
Height of half insulator	Cm
Main leakage path of half insulator	Cm
diameter (outside)	Cm
A.C.V withstand for 1min (dry)	kV
A.C.V withstand for 10 sec (wet)	kV
Dry flashover v	kV
Wet flashover v	kV
Impulse withstand v (1.2/50 μ s)	kV
Bending force	kN
Expulsion fuse int. capacity	
Sym	kA
Asym	kA

I/ We guarantee the technical data given above for the offered load break switch.

Signature:

Date: